

How k, c and m affect the building response

k	c	m	Expected response	Oscillation
1	1	1	Very large displacement; very fast and lively response	Strong
1	1	10	Very large displacement; low damping; longer period	Strong
1	1	100	Very large displacement; low damping; slow but strong motion	Strong
1	10	1	Large displacement; moderate damping; relatively fast	Moderate
1	10	10	Large displacement; moderate damping; noticeable oscillation	Moderate
1	10	100	Large displacement; moderate damping; very little oscillation	Moderate
1	100	1	Large motion but strongly damped	Weak
1	100	10	Large motion; strong damping; reduced oscillation	Weak
1	100	100	Large displacement; strong damping; slow response	Weak
10	1	1	Moderate-to-large motion; very little damping	Strong
10	1	10	Moderate motion; low damping; visible oscillation	Strong
10	1	100	Moderate motion; slow period; low damping	Strong
10	10	1	Moderate motion; reasonable damping	Moderate
10	10	10	Baseline case: balanced response	Moderate
10	10	100	Moderate response; slower motion	Moderate
10	100	1	Small displacement; strongly damped	Very weak
10	100	10	Reduced response; damped motion	Very weak
10	100	100	Small displacement; damped and slow	Very weak
100	1	1	Small displacement; weak damping; fast but not too large	Moderate
100	1	10	Small displacement; low damping; fast oscillation	Moderate
100	1	100	Small-to-moderate displacement; slow but stiff response	Moderate
100	10	1	Very small displacement; moderate damping	Weak
100	10	10	Small displacement; moderate damping	Weak